EECE 306 Lab 7 notebook. Writing Your Own Poisson Solver

Team 8. Thalea Collymore, Kroix Jones, Xavier Moore

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Setup

```
clear; close all;
e0 = em.const.eps0();

Vex = @(X,Y) sin(pi*X).*sin(pi*Y);
rho = em.test.mms2d(Vex, 1);
```

Manufactured solution and convergence

```
n = 101;
G = em.solve.grid2d([0 1], [0 1], n, n);
mask = false(n,n);
mask([1 end],:) = true;
mask(:,[1 end]) = true;

V = em.solve.poisson2d(G, rho, 1, mask, zeros(n,n), ...
    struct('method','direct'));

err101 = max(abs(V(:) - Vex(G.X(:),G.Y(:))));
fprintf('MMS max error on the 101 x 101 grid = %.3e\n', err101);

Ns = [21 41 81 161]';
errs = zeros(size(Ns));

for j = 1:numel(Ns)
    n = Ns(j);
    Gn = em.solve.grid2d([0 1], [0 1], n, n);
    mk = false(n,n);
    mk([1 end],:) = true;
    mk(:,[1 end]) = true;

    Vn = em.solve.poisson2d(Gn, rho, 1, mk, zeros(n,n), ...
        struct('method','direct'));

    errs(j) = max(abs(Vn(:) - Vex(Gn.X(:),Gn.Y(:))));
end

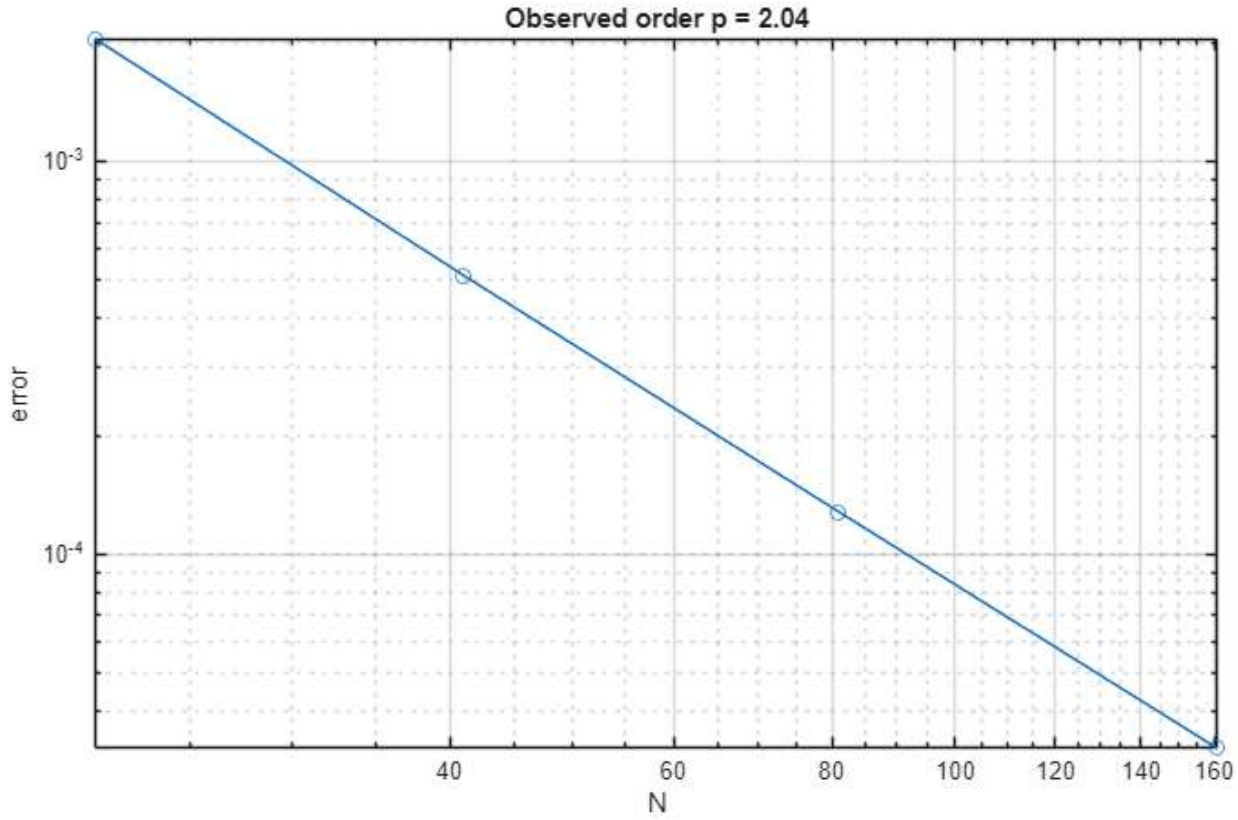
disp(table(Ns, errs, 'VariableNames', {'N','MaxError'}));

p = em.test.convergence(@(N) errs(Ns == N), Ns);
fprintf('observed order p = %.2f\n', p);
```

```
MMS max error on the 101 x 101 grid = 8.224e-05
     N      MaxError
    ___    __________

     21     0.0020587
     41    0.00051419
     81    0.00012851
    161    3.2116e-05

observed order p = 2.04
```



Three iterative methods

```
n = 41;
G = em.solve.grid2d([0 1], [0 1], n, n);
mask = false(n,n);
mask([1 end],:) = true;
mask(:,[1 end]) = true;

methods = {'jacobi','gs','sor'};

figure;
hold on

for k = 1:3
    opts = struct('method',methods{k}, ...
        'tol',1e-6,'maxit',20000);

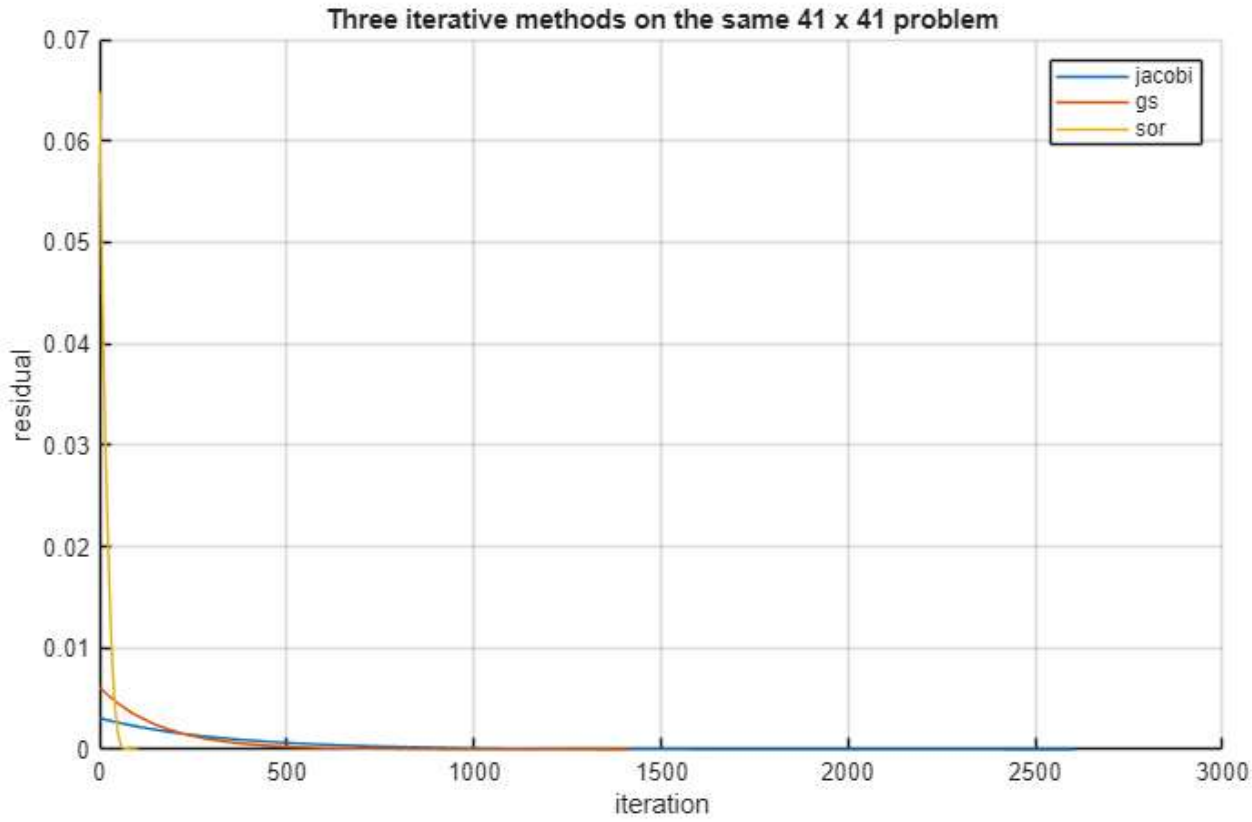
    if strcmp(methods{k},'sor')
        opts.omega = 2/(1+sin(pi/(n-1)));
    end
```

```
    [~,info] = em.solve.poisson2d(G, rho, 1, mask, ...
        zeros(n,n), opts);

    semilogy(info.resHist)
    fprintf('%-7s iterations = %d\n', ...
        methods{k}, info.iter);
end

grid on
legend(methods)
xlabel('iteration')
ylabel('residual')
title('Three iterative methods on the same 41 x 41 problem')
```

```
jacobi  iterations = 2604
gs      iterations = 1415
sor     iterations = 102
```



Omega scan on two grids

```
figure;
hold on

for n = [31 61]
    G = em.solve.grid2d([0 1], [0 1], n, n);
    mask = false(n,n);
    mask([1 end],:) = true;
    mask(:,[1 end]) = true;

    omega = linspace(1.0,1.99,20);
    iterations = zeros(size(omega));
```

```
    for j = 1:numel(omega)
        [~,info] = em.solve.poisson2d(G, rho, 1, mask, ...
            zeros(n,n), ...
            struct('method','sor','tol',1e-4, ...
            'maxit',20000,'omega',omega(j)));

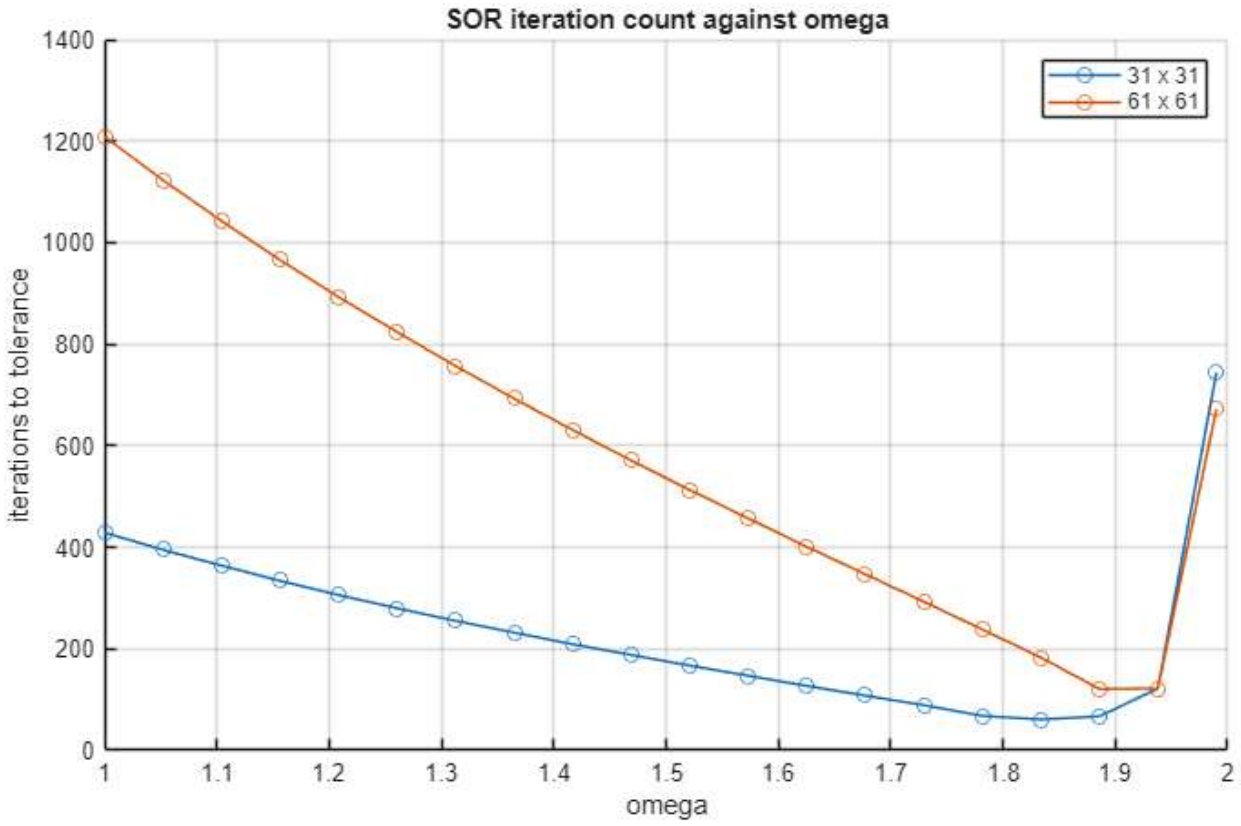
        iterations(j) = info.iter;
    end

    plot(omega,iterations,'o-')

    [~,best] = min(iterations);
    fprintf('n = %d   best omega about %.2f\n', ...
        n, omega(best));
end

grid on
legend('31 x 31','61 x 61')
xlabel('omega')
ylabel('iterations to tolerance')
title('SOR iteration count against omega')
```

```
n = 31   best omega about 1.83
n = 61   best omega about 1.89
```



Timing, iterative against direct

```
Nt = [41 61 81 101];
td = zeros(size(Nt));
ts = zeros(size(Nt));
```

```
for k = 1:numel(Nt)
    n = Nt(k);

    G = em.solve.grid2d([0 1], [0 1], n, n);
    mask = false(n,n);
    mask([1 end],:) = true;
    mask(:,[1 end]) = true;

    tic
    em.solve.poisson2d(G, rho, 1, mask, zeros(n,n), ...
        struct('method','direct'));
    td(k) = toc;

    tic
    em.solve.poisson2d(G, rho, 1, mask, zeros(n,n), ...
        struct('method','sor','tol',1e-6, ...
        'maxit',40000, ...
        'omega',2/(1+sin(pi/(n-1)))));
    ts(k) = toc;

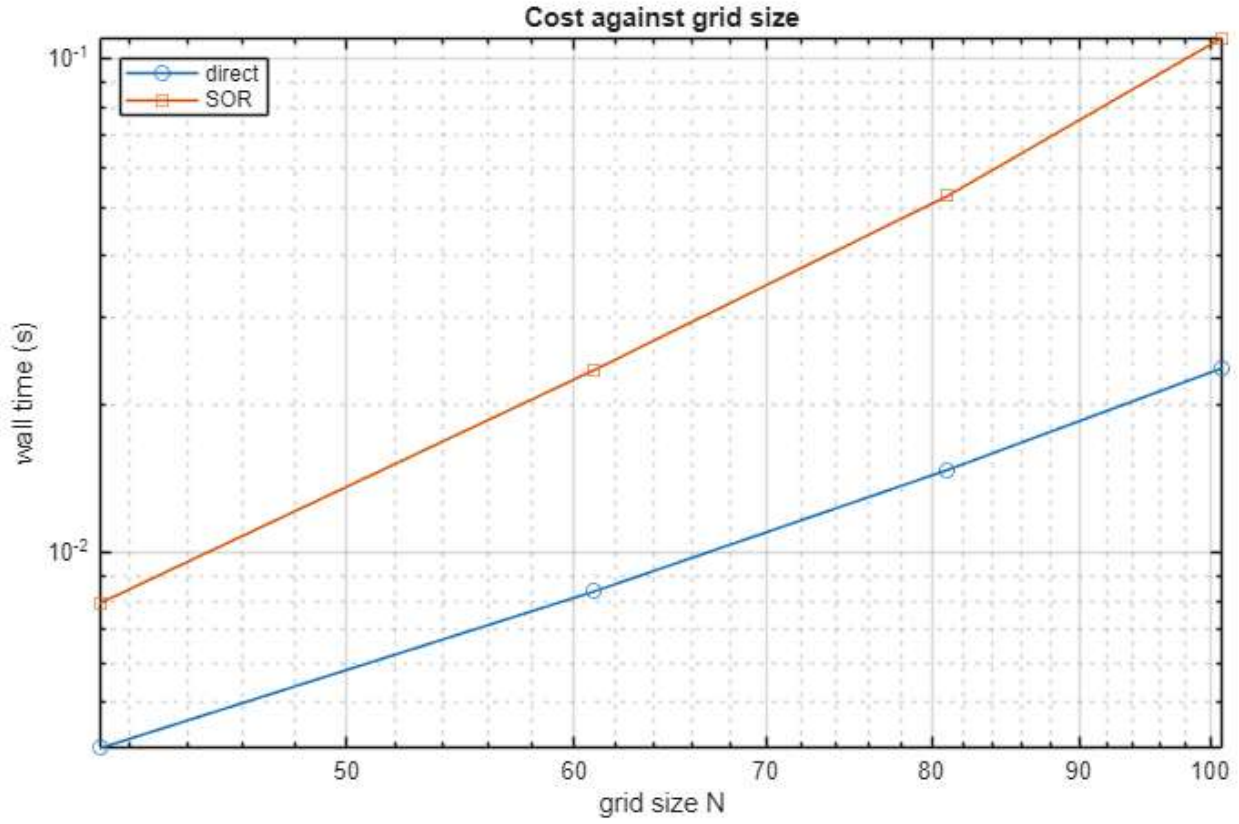
    fprintf('N = %d   direct %.3f s   SOR %.3f s\n', ...
        n, td(k), ts(k));
end

figure;
loglog(Nt,td,'o-',Nt,ts,'s-');
grid on
xlabel('grid size N')
ylabel('wall time (s)')
legend('direct','SOR','location','northwest')
title('Cost against grid size')

crossIdx = find(ts < td,1);

if isempty(crossIdx)
    fprintf('No timing crossover was observed through N = %d.\n', ...
        Nt(end));
else
    fprintf('First observed timing crossover is near N = %d.\n', ...
        Nt(crossIdx));
end
```

```
N = 41   direct 0.004 s   SOR 0.008 s
N = 61   direct 0.008 s   SOR 0.023 s
N = 81   direct 0.015 s   SOR 0.053 s
N = 101   direct 0.024 s   SOR 0.111 s
No timing crossover was observed through N = 101.
```



Interpretation

The manufactured solution converged at about second order, which matches the five-point stencil. If the measured order had been closer to one, the first things to check would be the boundary indexing and the Neumann boundary conditions. Gauss-Seidel needed fewer sweeps than Jacobi, while SOR was faster when omega was chosen near its best value. The best omega changed as the grid became finer, showing that one value does not work best for every grid.

Problems encountered

One problem we ran into was handling the zero-derivative boundary. At first, a missing neighbor outside the grid was replaced with the boundary node itself. The solution still looked smooth, so the mistake was not obvious. We corrected it by mirroring the interior neighbor across the boundary, which gives the centered zero-derivative condition required by the solver.

Full test suite

```
runTests
close all
```

```
EECE 306 toolbox test suite
------------------------------------------------------------

eta =

  376.7303

  test_lab01 checks complete
  PASS  test_lab01            0.02 s
  test_lab02 checks complete
  PASS  test_lab02            0.02 s
  5000-point field evaluation: 0.0004 s
  test_lab03 checks complete
  PASS  test_lab03            0.02 s
```

```
  test_lab04 checks complete
  PASS  test_lab04            0.05 s
  merged source: flux 5.00000000e-09 C, enclosed 5.00000000e-09 C
  density errors at 128000 / 1024000 elements: 0.953% / 1.715% max
  sphere / cube / closed hemisphere: 3.00000000e-09 / 3.00000000e-09 / 3.00000000e-09 C
  sphere Gauss rel err 1.44e-11; midpoint N=960 rel err 4.46e-07
  test_lab05 checks complete
  PASS  test_lab05            2.53 s
  Lab 6: max absolute map error 7.776e-04 V/m
  max closed residual / (max|E| L) 2.473e-13
  Square vs exact-tangent edges 4.178e-11 V
  open midpoint errors [0.0004329 0.0001082 2.706e-05 6.764e-06] V (order 2)
  Fixed line refinement 4.965e-06 V
  growing line potentials 25.9495 -> 37.6527 V
  warning verified
  Step minimum h=3.162e-06 m, error 1.025e-11
  default error
  10000 points V/grad 0.001/0.004 s
  test_lab06 checks complete
  PASS  test_lab06            1.10 s
  Lab 7: MMS error 8.224e-05, order 2.042
  Jacobi / GS / SOR iterations 1320 / 689 / 77
  Gauss flux 2.5947e-09 C, enclosed 2.5000e-09 C
  test_lab07 checks complete
  PASS  test_lab07            0.30 s
 ----------------------------------------------------------------
  7 passed, 0 failed, 7 total
  All tests passing.
```

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